CBSE Class 10 Mathematics - Answer Key

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1. Which of the following equations is a quadratic equation?

Answer: (C)

Solution:

1. Check each option by expanding and simplifying.
2. Option (A) simplifies to $x^2 + 1 = x^2 + 2x + 1 \Rightarrow 2x = 0$, which is linear.
3. Option (B) simplifies to $x^2 + 2x + 1 = x^2 + 2x$, which is not an equation.
4. Option (C) is $x^2 - 3x = 2$, which is in the form $ax^2 + bx + c = 0$.
5. Option (D) simplifies to $x^3 = x^2 + 1$, which is a cubic equation.

$$x^2 - 3x = 2$$
$$\Rightarrow x^2 - 3x - 2 = 0$$

2. The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is

Answer: (B)

Solution:

1. Identify coefficients $a = 2, b = -4, c = 3$.
2. Use the formula $D = b^2 - 4ac$.
3. Substitute values: $D = (-4)^2 - 4(2)(3)$.
4. Calculate: $D = 16 - 24 = -8$.

$$D = (-4)^2 - 4 \times 2 \times 3$$
$$D = 16 - 24$$
$$D = -8$$

3. If $x = 3$ is a root of the equation $x^2 - kx - 6 = 0$, then the value of k is

Answer: (A)

Solution:

1. Since $x = 3$ is a root, it must satisfy the equation.
2. Substitute $x = 3$ into the equation: $(3)^2 - k(3) - 6 = 0$.
3. Solve for k: $9 - 3k - 6 = 0$.
4. $3 - 3k = 0 \Rightarrow 3k = 3 \Rightarrow k = 1$.

$$
\begin{aligned}
3^2 - 3k - 6 &= 0 \\
9 - 6 &= 3k \\
3 &= 3k \\
k &= 1
\end{aligned}
$$

4. For what value of p does the quadratic equation $x^2 + px + 9 = 0$ have equal roots?

Answer: (C)

Solution:

1. For equal roots, the discriminant $D = b^2 - 4ac$ must be 0.
2. Here $a = 1, b = p, c = 9$.
3. Set $D = p^2 - 4(1)(9) = 0$.
4. $p^2 - 36 = 0 \Rightarrow p^2 = 36$.
5. Taking square root, $p = \pm 6$.

$$
\begin{aligned}
p^2 - 4(1)(9) &= 0 \\
p^2 &= 36 \\
p &= \pm 6
\end{aligned}
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